

Multicolumn Exponential Periods: Conditional Comparison and Geometric Realization

Christopher D. Long

Antoine-Auguste Le Blanc*

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galizur@gmail.com

Abstract

This candidate draft isolates the algebraic orbit calculations useful for multicolumn exponential-period comparison and distinguishes them from their unproved realization by actual collections of chains. We prove the rectangular and square special-linear orbit statements, explain joint determinant-character relations, and give an explicit conditional descent criterion. The missing geometric and arithmetic hypotheses are listed rather than assigned to a formal period category by definition. No unrestricted multichain numerical comparison theorem is asserted.

Candidate research draft; not a numbered paper in the series.

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1 Why a single column is different

Paper V [1] studies a chosen actual vector U satisfying $U' = AU + b$ over a fixed boundary field. An affine differential Galois group can have a full translation kernel, which gives a dense affine orbit. If the kernel is zero, a reductive cocycle splits, giving

$$U = a + Yc,$$

*Le Blanc cryptographic author identifier: b604812530b3535b23d6dbb0f5abe32d1ed1a9de35fb4f7d04d62cff08ea19c8

where Y is a fundamental matrix and c is a constant vector. For a homogeneous group containing SL_n , one nonzero vector still has dense orbit when $n \geq 2$.

Several chains produce several columns. Even when every individual column passes the one-column test, their combined orbit can satisfy determinant relations. Thus a product of individually nonzero scalar periods, as used in Paper VI [2], is not a multicolumn independence theorem.

2 Explicit special-linear orbits

Let C be an algebraically closed field of characteristic zero, let $n \geq 2$, and let $\mathrm{SL}_n(C)$ act on $n \times r$ matrices by left multiplication.

Proposition 2.1 (Rectangular columns). *If $1 \leq r < n$ and a matrix Z has column rank r , its SL_n orbit is the set of all rank- r matrices. This orbit is Zariski dense in $\mathbb{A}^{nr}$ and has zero polynomial vanishing ideal.*

Proof. Given two ordered independent r -tuples, extend them to bases of C^n . There is a change of basis carrying one tuple to the other. Because an unused basis vector exists, rescale its image to make the determinant one without changing the prescribed first r columns. The full-rank locus is a nonempty open subset of affine space, hence dense. $\square$

Proposition 2.2 (Square columns). *If $r = n$ and Z is invertible, its orbit is*

$$\{X : \det X = \det Z\}.$$

The polynomial ideal of the orbit is generated by $\det X - \det Z$.

Proof. For any X with the same nonzero determinant, $XZ^{-1} \in \mathrm{SL}_n$. The fiber is a translate of SL_n , hence irreducible and reduced. For completeness, the differential of the determinant at an invertible matrix is nonzero, so the fiber is smooth. Its defining principal ideal is therefore radical, and the displayed set equality gives the ideal. $\square$

For example, two independent columns for SL_2 have the determinant invariant $x_{11}x_{22} - x_{12}x_{21}$, although either column alone has dense orbit in $\mathbb{A}^2$. This example is an algebraic group calculation, not a realized pair of exponential chains.

Remark 2.3 (Dependent and excess columns). If the actual constant column matrix has rank less than r , its linear column relations must be imposed first. If $r > n$, maximal minors and their relations enter; rectangular density cannot be applied without a rank reduction. These cases require the correct actual constant matrix, not a chosen formal substitute.

3 Several blocks and joint determinant characters

Suppose distinct homogeneous blocks have derived groups containing a product of special-linear groups. For a full-rank square block s , put $\delta_s = \det Z_s$. Its residual transformation is governed by a character χ_s of a common algebraic group G .

Proposition 3.1 (Joint character equations). *For the morphism*

$$\chi : G \longrightarrow (\mathbb{G}_m)^b, \quad g \longmapsto (\chi_1(g), \dots, \chi_b(g)),$$

let $T = \chi(G)$ and $\Lambda = \{a \in \mathbb{Z}^b : \chi_1^{a_1} \cdots \chi_b^{a_b} = 1\}$. The Laurent ideal of the orbit δT is

$$(X^a - \delta^a : a \in \Lambda) \subset C[X_1^{\pm 1}, \dots, X_b^{\pm 1}].$$

Proof. A closed subgroup of a torus is defined by the characters trivial on it. Equivalently its coordinate algebra is the group algebra $C[\mathbb{Z}^b/\Lambda]$, and the quotient map on Laurent monomials has kernel generated by $X^a - 1$, $a \in \Lambda$. Translation by δ gives the formula. This also covers a disconnected image: in characteristic zero the finite diagonalizable part is reduced. $\square$

Independent determinant equations for the separate blocks need not suffice. For example the image $\{(t, t) : t \in C^\times\}$ has neither coordinate fixed but has the joint equation $X_1\delta_2 - X_2\delta_1 = 0$.

The archived pre-single-chain manuscript [3] develops a more general affine-orbit calculation, including active translation columns, reductive splitting and local determinant-character models. The present note does not reclassify arbitrary isotypic affine extensions. In particular the hypothesis that a derived group is a product of independent special-linear factors must be proved for the actual blocks.

4 A conditional actual specialization criterion

The following algebraic formulation isolates what is required after an actual functional comparison has been made. A discrete valuation ring is denoted by R , with uniformizer π , fraction field K , and residue field κ .

Proposition 4.1 (Local contraction and specialization). *Let $I \subset R[X_1, \dots, X_N]$ and suppose $A = R[X]/I$ is R -flat. Then*

$$I = (IK[X]) \cap R[X].$$

Suppose further that an actual functional period presentation has generic kernel $IK[X]$, and that every numerical relation at the chosen parameter lifts to a functional relation with coefficients in R and the prescribed residue row or polynomial. Then its numerical kernel is exactly the specialization of I , provided every specialized generator of I is an actual numerical relation.

Proof. Flatness implies torsion-freeness over the discrete valuation ring. If the class of $P \in R[X]$ vanishes after tensoring with K , it is annihilated by a power of π , hence already zero. This proves contraction. A lifted numerical relation is in the generic kernel and therefore in I by contraction; reducing modulo π proves one kernel inclusion. The stipulated validity of specialized generators proves the other inclusion. $\square$

The assertion is conditional on the actual presentation and the lifting hypothesis. It does not prove those hypotheses. In a period application, R must be chosen from actual boundary coefficients regular at the specified parameter; a formal localization at unspecified denominators is insufficient. Beukers' lifting theorem [4] applies to closed E -function systems at nonsingular nonzero algebraic arguments, not automatically to an arbitrary formal coefficient ring.

5 What a genuine multichain theorem must verify

A publishable realization should exhibit specific chains and prove the following items in dependency order.

1. *Actual columns and common base.* Derive the common inhomogeneous system by literal polynomial Stokes certificates, and identify all actual linear relations among columns. Passing to a cover cannot silently be assumed to preserve irreducibility or independence of blocks.

2. *Homogeneous and affine groups.* Verify the product derived group, its surviving boundary-field action, and the full translation subspace. Isomorphic blocks require an isotypic analysis, not separate copies of an irreducible representation by fiat.
3. *Determinant realization.* Construct determinant-line relations with the actual period normalizations. An invariant of an abstract fundamental matrix does not by itself identify a formal Stokes relation between the prescribed chains.
4. *Joint characters and local units.* Compute the entire character lattice and prove that determinant coefficients and affine shifts have the required regularity and nonvanishing at specialization.
5. *Numerical lifting.* Place the actual coordinates in a closed E -function system and verify the ordinary-point hypotheses and local contraction used in Proposition 4.1.

The earlier unrestricted endpoint-complete claim in the development history was withdrawn. Neither this candidate nor the new scalar theorem in Paper V reinstates it.

6 A concrete first target and stopping rule

The smallest informative example is a rank-two interior system with two geometrically different compact chains sharing a specified boundary system. A successful example should first determine whether the affine translation kernel makes both columns free or whether a normalized determinant survives. In the latter case the determinant must be computed as an actual boundary quantity and checked at every claimed algebraic scale. A symbolic 2×2 determinant calculation is only the orbit part of this task.

The candidate is deliberately not assigned a series number. Its proved statements are the elementary orbit and local-contraction propositions above. The missing theorem is an actual multichain realization satisfying all five requirements, with at least one fully checked example. The archived proofs and the current V should be cited at their exact scope, not used as evidence that the realization has already been completed.

References

- [1] C. D. Long and A.-A. Le Blanc, *Relative Exponential Periods on Polynomial Chains: Algebraic Independence over Fixed Boundary Fields*, Paper V, September 2026, revised October 2026.
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